

Homework 1 Reflection:

What AI-Driven Mathematics Means for My Future in FinTech

Name: YANG, Yahan

Student ID: 1155272721

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On 23 September 2026, researchers released a preprint titled *Learning to Discover Interesting Mathematics*. The study explores how AI can generate mathematical conjectures and select promising ones for further investigation, rather than only answer questions provided by humans. The authors propose a measurable criterion for identifying interesting theorems and describe a system that builds on a growing mathematical library. Although this is a preprint rather than a settled scientific conclusion, it made me reconsider the role AI could play in analytical work.

My initial reaction was a mixture of excitement and concern. As a finance graduate now studying financial technology, I have been developing programming and modelling skills partly because I believed they would provide a durable professional advantage. Seeing AI move towards mathematical discovery challenged the assumption that technically difficult work would necessarily remain a human strength. However, the paper also raised a question that I found more useful than asking whether AI would replace analysts: who decides which problems are worth solving? A theorem can satisfy a mathematical criterion without being useful for a particular application. Similarly, a financial model can achieve strong predictive performance while addressing the wrong business question.

This distinction became clearer through our receipt-analysis homework. My program needed to distinguish the final payment from the subtotal, add back discounts, and exclude rounding adjustments from those discounts. These sound like small details, but they determine what the answers actually mean. The model extracted information, while Python performed the calculations. Both aggregate answers were correct in one public test, but this did not establish that the program would handle every unfamiliar receipt correctly. I learned that a successful demonstration is only one part of building a reliable system. Clear definitions, appropriate tests and careful interpretation remain necessary.

My internships in banking, securities and trust businesses give this lesson a practical context. When reviewing company information, I had to compare financial statements with other documents and investigate inconsistencies. AI could help with extraction and preliminary comparison, but deciding whether two figures are comparable requires knowledge of reporting periods, accounting definitions and the underlying business. I therefore see an opportunity to combine financial understanding with the ability to design and evaluate AI-assisted workflows.

This event has made my career plan more specific. I still want to pursue analytical or technology-related roles in finance, but I will focus less on learning models as isolated techniques. My next step is to build a small financial-document analysis project that records its sources, flags inconsistencies and requires human review for uncertain results. Alongside Python and API skills, I want to strengthen my understanding of statistics and model evaluation. My goal is to become someone who can choose useful problems, work effectively with AI and explain why the resulting analysis deserves trust.

Reference:

Patel, N., Rammal, A., Hayat, A., Munos, R., & Kempe, J. (2026). *Learning to Discover Interesting Mathematics*. arXiv preprint, submitted 23 September 2026.

<https://arxiv.org/abs/2609.28603>